

Activity 2 - Data Visualization using Matplotlib

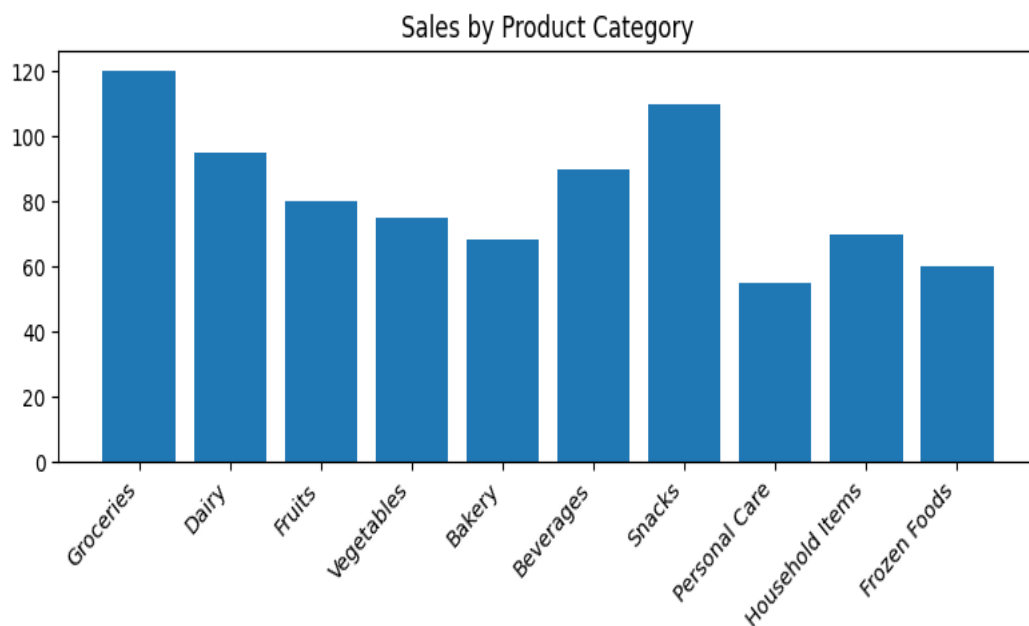
Question 1 - Bar Chart

Dataset

Groceries: 120 Lakhs
Dairy: 95 Lakhs
Fruits: 80 Lakhs
Vegetables: 75 Lakhs
Bakery: 68 Lakhs
Beverages: 90 Lakhs
Snacks: 110 Lakhs
Personal Care: 55 Lakhs
Household Items: 70 Lakhs
Frozen Foods: 60 Lakhs

Python Code

```
import matplotlib.pyplot as plt
categories=[...]
sales=[120,95,80,75,68,90,110,55,70,60]
plt.bar(categories,sales)
plt.show()
```



Analysis

Highest: Groceries (120 Lakhs)
Lowest: Personal Care (55 Lakhs).
Groceries perform best while Personal Care performs least.

Question 2 - Histogram

Dataset

3.2, 4.5, 5.0, 6.8, 7.2, 8.5, 4.8, 5.5, 6.0, 7.8, 9.2, 10.5

Python Code

```
plt.hist(salary, bins=5)  
plt.show()
```



Analysis

Range: 3.2–10.5 LPA. Most salaries fall between 4–8 LPA. 10.5 LPA is a possible high outlier.

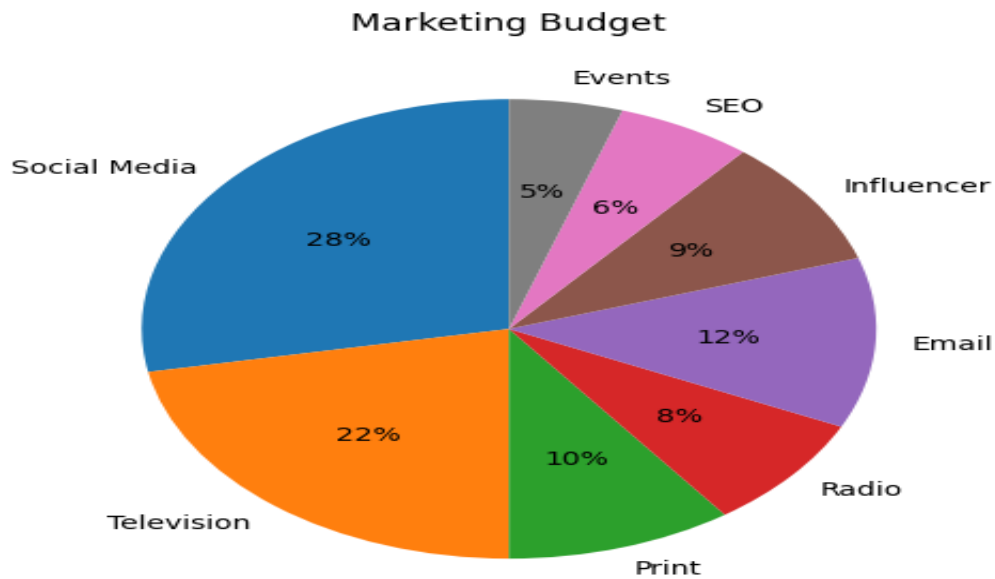
Question 3 - Pie Chart

Dataset

Social Media: 28%
Television: 22%
Print: 10%
Radio: 8%
Email: 12%
Influencer: 9%
SEO: 6%
Events: 5%

Python Code

```
plt.pie(budget, labels=channels, autopct='%1.1f%%')
```



Analysis

Highest allocation: Social Media (28%). Budget is mainly concentrated in digital marketing.

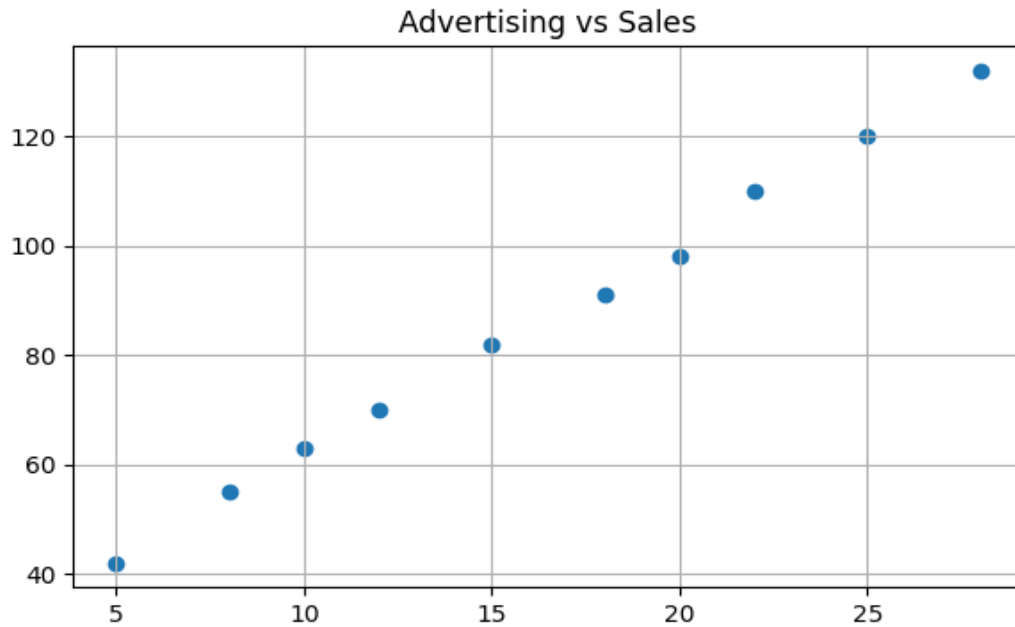
Question 4 - Scatter Plot

Dataset

5 Lakhs -> 42 Lakhs
8 Lakhs -> 55 Lakhs
10 Lakhs -> 63 Lakhs
12 Lakhs -> 70 Lakhs
15 Lakhs -> 82 Lakhs
18 Lakhs -> 91 Lakhs
20 Lakhs -> 98 Lakhs
22 Lakhs -> 110 Lakhs
25 Lakhs -> 120 Lakhs
28 Lakhs -> 132 Lakhs

Python Code

```
plt.scatter(advertising,sales)
```



Analysis

Strong positive relationship: higher advertising cost leads to higher sales revenue.

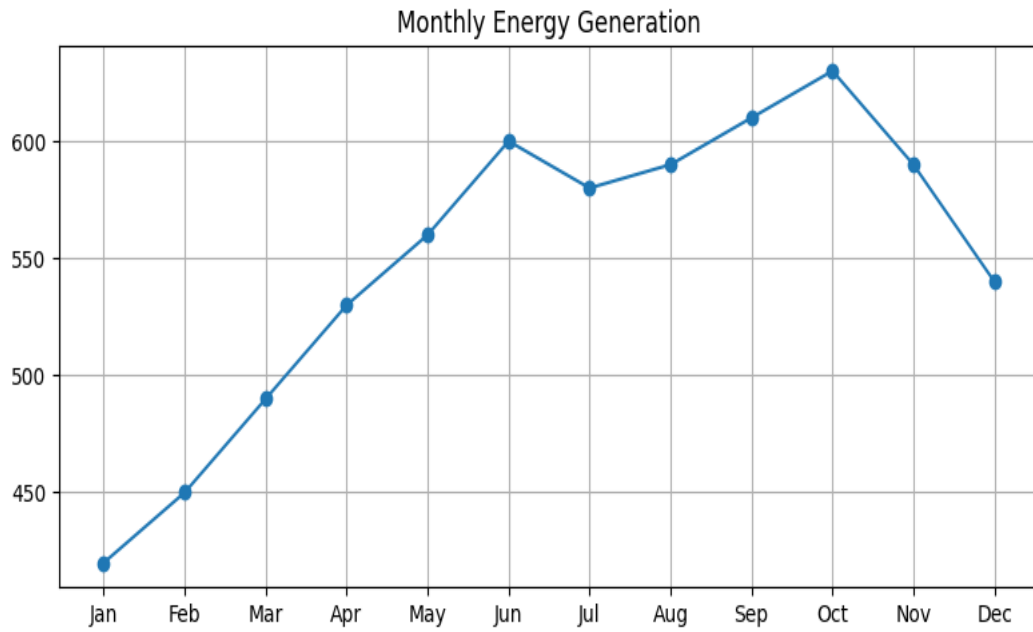
Question 5 - Line Chart

Dataset

Jan: 420 MWh
Feb: 450 MWh
Mar: 490 MWh
Apr: 530 MWh
May: 560 MWh
Jun: 600 MWh
Jul: 580 MWh
Aug: 590 MWh
Sep: 610 MWh
Oct: 630 MWh
Nov: 590 MWh
Dec: 540 MWh

Python Code

```
plt.plot(months,energy,marker='o')
```



Analysis

Highest: October (630 MWh). Lowest: January (420 MWh). Trend increases overall with slight dips.